

NYX

Operational Decisioning Platform

ENTERPRISE OVERVIEW & OPERATIONAL VISION

Obsidian Defense Systems

Table of Contents

EXECUTIVE SUMMARY 3

THE PROBLEM SPACE..... 3

TIME TO DECISION (TTD) 5

USE CASES 6

 USE CASE 1: EDR ALERT TRIAGE 6

 USE CASE 2: VULNERABILITY PRIORITIZATION..... 7

INTEGRATION MODEL..... 9

OPERATIONAL GOVERNANCE & HUMAN REVIEW 11

SECURITY & GOVERNANCE 13

STRATEGIC OUTLOOK..... 15

APPENDIX..... 17

 APPENDIX A – NYX PLATFORM DIFFERENTIATION 17

 APPENDIX B – TRINITY DECISION MODEL 18

 APPENDIX F – EXAMPLE INTEGRATION ARCHITECTURE 19

EXECUTIVE SUMMARY

Modern cybersecurity operations are no longer limited by detection capability. They are limited by the ability to rapidly interpret, prioritize, and act on an overwhelming volume of security data.

Today's enterprise environments generate thousands of alerts daily across SIEM, EDR, vulnerability management, monitoring, identity, and threat intelligence platforms. While detection technologies have significantly matured, security operations teams continue to face growing challenges surrounding alert fatigue, analyst overload, inconsistent triage outcomes, and delayed response times.

NYX was designed to address this operational gap.

NYX is a cybersecurity decision engine that operates alongside existing security technologies, ingesting and correlating data across multiple security and infrastructure domains to produce structured, defensible decisions at machine speed.

At its core, NYX utilizes a consensus-based reasoning model built around three operational roles:

- ➡ Analyst
- ➡ Judge
- ➡ Auditor

Together, these components form the Trinity Decision Model, enabling NYX to evaluate context, challenge assumptions, validate outcomes, and generate auditable decisions while maintaining human oversight where required.

NYX is designed to reduce low-value analyst workload by automating and validating repetitive decision processes surrounding low, informational, and select medium-severity events. Simultaneously, it enhances operational visibility by enriching alerts with contextual intelligence gathered from multiple integrated platforms before delivering actionable outputs into existing workflows and ticketing systems.

The result is improved operational consistency, reduced noise, faster Time to Decision (TTD), lower false positive impact, and a more scalable approach to modern cybersecurity operations.

THE PROBLEM SPACE

Modern cybersecurity operations centers are facing a scalability crisis.

Enterprise environments now generate massive volumes of telemetry and alerts across SIEM, EDR, vulnerability management, identity, cloud, monitoring, and threat intelligence platforms. While organizations have significantly expanded their detection capabilities over the last decade, operational decision-making has not evolved at the same pace.

The result is an increasingly fragmented and overloaded security workflow.

Security teams today must contend with:

- High alert volume across disconnected platforms
- Manual triage processes that do not scale

- Inconsistent decision-making between analysts and teams
- Growing analyst fatigue and operational burnout
- Delayed response times caused by human bottlenecks
- Limited contextual visibility across security domains
- Insufficient auditability surrounding operational decisions

As organizations continue adding new security technologies, many inadvertently increase operational complexity rather than reduce it. More telemetry does not automatically translate into better security outcomes.

This challenge is further amplified by staffing shortages, increasing infrastructure scale, and the growing requirement for rapid, defensible decision-making across modern enterprise environments.

The issue is no longer detection capability. The issue is operational decision making at scale.

NYX OVERVIEW

NYX operates as a cybersecurity decision engine and operational intelligence layer within an organization's existing security architecture.

Rather than replacing SIEM, EDR, vulnerability management, monitoring, or ticketing platforms, NYX is designed to operate alongside them by ingesting, correlating, enriching, and evaluating operational data across multiple security domains.

This approach allows organizations to augment existing security investments without requiring disruptive rip-and-replace initiatives or major workflow restructuring. Unlike traditional SOAR platforms, which primarily focus on workflow orchestration and task automation, NYX is designed to address the decision-making gap that exists between alert generation and operational response.

NYX focuses on:

- Interpreting context across multiple security domains
- Producing structured, defensible decisions
- Reducing repetitive analyst-driven triage
- Enriching operational workflows with correlated intelligence
- Improving consistency and scalability within security operations

At the core of NYX is the Trinity Decision Model, a consensus-based reasoning framework consisting of three operational roles:

- Analyst
- Judge
- Auditor

Together, these components evaluate incoming events, validate reasoning, challenge assumptions, and generate auditable decision outputs designed to support both operational speed and governance requirements.

Core platform capabilities include:

- Multi-source alert ingestion
- Cross-domain contextual enrichment
- Consensus-based decisioning
- Human review and approval workflows
- Audit-ready operational outputs
- Optional policy-driven automation capabilities

Automation within NYX is intentionally configurable and governed by organizational policy. Out-of-the-box deployments are designed to support human oversight first, with optional automation and remediation workflows enabled selectively based on operational maturity and risk tolerance.

By operating as a parallel intelligence and decision layer, NYX enables organizations to improve operational efficiency, reduce alert fatigue, lower false positive impact, and accelerate Time to Decision (TTD) without replacing the security technologies already in place.

TIME TO DECISION (TTD)

NYX introduces a foundational operational metric known as Time to Decision (TTD). TTD measures the time required to progress from alert ingestion to a structured, defensible, operational decision.

While traditional cybersecurity metrics such as Mean Time to Detect (MTTD) or Mean Time to Respond (MTTR) remain valuable indicators of operational performance, they primarily measure detection and response activities after a workflow has already been initiated. TTD was designed to address the increasingly critical gap between detection and action.

As modern enterprise environments continue generating massive volumes of telemetry and alerts, the ability to rapidly and consistently interpret operational context has become a primary scalability challenge for cybersecurity teams. In many environments, the true bottleneck is no longer identifying potential threats, but determining which actions should be taken, when they should occur, and whether those decisions can be operationally defended.

TTD provides a measurable framework for evaluating cybersecurity decision efficiency at scale.

NYX is architected to reduce operational decision latency through consensus-based reasoning, contextual enrichment, and parallelized analysis workflows. Under optimized deployment conditions, NYX is designed to support sub-second operational decisioning for eligible workflows while maintaining human oversight capabilities where required.

Unlike traditional response metrics, TTD can be segmented into multiple operational categories to provide greater visibility into workflow performance and decision pipeline efficiency.

Core TTD Categories

- **Machine TTD**
Measures automated decision latency without human intervention.
- **Trinity Engine TTD**
Measures the time required for the Analyst, Judge, and Auditor components to complete consensus evaluation.

- **AI TTD**
Measures latency introduced by AI-assisted reasoning and enrichment processes.
- **Human Review TTD**
Measures the time between escalation into human review and human analyst approval or override.
- **End-to-End TTD**
Measures total operational decision time from alert ingestion through final action or escalation.

By breaking decision workflows into measurable operational components, organizations gain visibility into where delays occur, how automation impacts efficiency, and where operational bottlenecks continue to exist.

TTD is intended to serve not only as a performance metric, but as a strategic framework for measuring cybersecurity operational maturity in environments increasingly constrained by alert volume, analyst workload, and decision complexity.

USE CASES

USE CASE 1: EDR ALERT TRIAGE

Environment

- CrowdStrike
- SentinelOne
- Microsoft Defender for Endpoint
- Similar EDR/XDR platforms

Operational Challenge

Modern EDR platforms generate large volumes of behavioral detections and security telemetry, creating operational strain for security teams responsible for triage and response.

Organizations commonly face:

- High alert volume
- Duplicate or overlapping detections
- Manual enrichment workflows
- Inconsistent triage outcomes
- Excessive false positive investigation
- Analyst fatigue and operational burnout

As enterprise environments scale, these challenges increasingly impact response efficiency and operational consistency.

NYX Approach

NYX operates alongside existing EDR platforms as a decision intelligence layer.

The platform ingests and enriches EDR detections using contextual intelligence such as:

- Asset criticality
- Vulnerability exposure
- Identity context
- Threat intelligence
- Historical activity
- Environmental risk indicators

Using the TRINITY consensus framework, NYX evaluates detections through:

- Analyst interpretation
- Judge validation
- Auditor verification

This process enables structured, explainable, and auditable operational decisioning.

Example Outcomes

NYX may:

- Prioritize high-risk activity
- Consolidate related detections
- Reduce duplicate investigations
- Recommend remediation actions
- Route events into human review workflows
- Enrich downstream ServiceNow or incident workflows

Optional policy-driven automation can be enabled where organizationally appropriate.

Operational Value

NYX helps organizations:

- Reduce repetitive analyst workload
- Improve triage consistency
- Lower false positive impact
- Accelerate Time to Decision (TTD)
- Improve signal-to-noise ratio
- Strengthen operational governance and auditability

Rather than replacing EDR platforms, NYX augments existing detection investments through contextual enrichment and structured operational decisioning.

USE CASE 2: VULNERABILITY PRIORITIZATION

Environment

- Qualys
 - Tenable
 - Rapid7
 - Similar vulnerability management platforms
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Operational Challenge

Modern enterprise environments generate massive volumes of vulnerability findings across infrastructure, cloud, endpoint, and application assets.

While vulnerability management platforms provide strong visibility into technical exposure, prioritization often relies heavily on static severity scoring that lacks sufficient operational and business context.

Organizations commonly face:

- Large remediation backlogs
- Severity inflation across findings
- Inconsistent remediation prioritization
- Limited visibility into actual operational risk
- Repetitive remediation effort
- Difficulty aligning remediation to business impact

In many environments, vulnerabilities with identical CVSS scores may represent significantly different levels of real-world operational risk.

NYX Approach

NYX operates alongside existing vulnerability management platforms as a contextual decision intelligence layer.

The platform enriches findings using operational intelligence such as:

- Asset criticality
- Internet exposure
- Threat intelligence
- Known exploit activity
- Business ownership
- Compensating controls
- Environmental dependencies

Using the TRINITY consensus framework, NYX evaluates vulnerabilities through:

- Analyst interpretation
- Judge prioritization validation
- Auditor verification

This enables structured, risk-informed remediation prioritization beyond static severity scoring alone.

Example Outcomes

NYX may:

- Prioritize vulnerabilities based on operational risk
- Recommend remediation timelines

- Support risk acceptance workflows
- Enrich ServiceNow or JIRA remediation tickets
- Improve remediation queue prioritization
- Reduce duplicate remediation effort

Recommendations remain configurable and aligned to organizational governance and risk tolerance.

Operational Value

NYX helps organizations:

- Reduce remediation backlog noise
- Improve prioritization consistency
- Focus remediation efforts on meaningful operational risk
- Accelerate remediation decision workflows
- Improve governance and auditability
- Align vulnerability management more closely to business impact

Rather than replacing vulnerability management platforms, NYX augments existing workflows through contextual enrichment, consensus-based reasoning, and structured operational decisioning.

INTEGRATION MODEL

NYX is designed to operate within existing enterprise security architectures through a flexible integration model focused on interoperability, operational visibility, and minimal disruption to established workflows.

Rather than requiring organizations to replace existing technologies, NYX is architected to integrate alongside current security, infrastructure, and operational platforms as a parallel intelligence and decision layer. This approach enables organizations to augment existing security investments while preserving operational continuity and architectural flexibility.

Supported Ingestion Methods

NYX supports multiple ingestion and integration mechanisms, including:

- REST APIs
- Webhooks
- Event stream ingestion
- Batch import workflows
- Structured data ingestion pipelines
- Log and telemetry forwarding integrations

This flexible architecture allows NYX to adapt to a wide range of enterprise environments, including hybrid, legacy, cloud-enabled, and highly regulated operational infrastructures.

Supported Integration Domains

NYX is designed to integrate across multiple operational and security domains, including:

- SIEM platforms
- EDR/XDR platforms
- Vulnerability management solutions
- Infrastructure and observability platforms
- Identity and access management systems
- Threat intelligence platforms
- Ticketing and workflow management systems
- Cloud and hybrid infrastructure environments

Supported technologies may include, but are not limited to:

- Splunk
- Microsoft Sentinel
- CrowdStrike
- Qualys
- LogicMonitor
- ServiceNow

Additional integrations may be supported through API connectivity, custom ingestion pipelines, or organizational integration workflows.

Deployment Flexibility

NYX is architected primarily for on-premise and controlled-environment deployments where operational visibility, governance, and data control requirements are critical.

Supported deployment models may include:

- On-premise deployment
- Air-gapped environments
- Hybrid enterprise infrastructure
- Private cloud environments
- Controlled cloud-enabled deployments

While cloud deployment models may be supported where operationally appropriate, NYX is intentionally designed to function without dependency on external cloud AI services or persistent internet connectivity.

This approach enables organizations to maintain greater control over:

- Operational data
 - Decision workflows
 - Security governance
 - Infrastructure segmentation
 - Regulatory and compliance requirements
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Operational Workflow Integration

NYX is designed to integrate into existing operational workflows rather than replace them.

Depending on organizational requirements, NYX may:

- Enrich existing ServiceNow incidents
- Support prioritization workflows
- Append contextual intelligence to operational tickets
- Route decisions into human review queues
- Trigger optional policy-driven automation workflows
- Provide centralized operational decision visibility

This integration model allows organizations to incrementally adopt advanced operational decisioning capabilities while maintaining existing governance structures and analyst oversight.

OPERATIONAL GOVERNANCE & HUMAN REVIEW

NYX is designed around a human-governed operational model that prioritizes trust, auditability, and controlled automation within enterprise security environments.

Rather than pursuing unrestricted autonomous response, NYX enables organizations to progressively scale operational automation based on policy maturity, organizational risk tolerance, and operational confidence.

This approach allows organizations to maintain human oversight where required while incrementally reducing repetitive analyst-driven decision workloads.

Governance-Driven Decisioning

NYX supports configurable operational governance controls, including:

- Role-based approval workflows
- Human review routing
- Configurable automation permissions
- Audit and decision traceability

These controls enable organizations to tailor operational decision workflows to align with internal governance requirements, compliance obligations, and security maturity levels.

Human Review Workflows

Where operational confidence thresholds are not met, NYX may route decisions into a Human Review Queue for analyst validation, escalation, or override.

Examples may include:

- High-risk operational actions
- Low-confidence decision outcomes
- Sensitive infrastructure systems
- Critical business applications

- Policy-defined escalation scenarios

Human review workflows are designed to ensure that operational oversight remains embedded within the decision lifecycle while still enabling accelerated triage and contextual enrichment.

Tiered Operational Automation

NYX supports multiple operational decision modes designed to align automation capabilities with organizational risk tolerance.

Low / Informational Events

Organizations may optionally configure NYX to:

- Auto-clear known low-value alerts
 - Suppress repetitive false positives
 - Apply informational disposition tagging
 - Perform low-risk workflow enrichment actions
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Medium-Severity Events

Organizations may configure NYX to:

- Recommend remediation actions
 - Route events into approval workflows
 - Enrich operational tickets
 - Trigger conditional automation workflows
 - Escalate based on contextual risk evaluation
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High / Critical Events

High-risk operational actions are intentionally governed and disabled by default unless explicitly authorized by organizational policy.

Where operationally appropriate, organizations may choose to enable advanced response capabilities such as:

- Automated containment workflows
- Host isolation
- Identity protection actions
- Emergency remediation procedures
- High-priority escalation workflows

These capabilities remain fully configurable and subject to organizational governance controls, approval models, and operational trust requirements.

Auditability & Operational Transparency

All NYX decisions, recommendations, approvals, overrides, and automated actions are designed to generate auditable operational records.

This includes:

- Decision rationale
- Trinity evaluation outputs
- Confidence scoring
- Human approvals and overrides
- Automation execution history
- Operational timestamps and TTD metrics

This model enables organizations to maintain operational transparency while improving consistency and scalability across security workflows.

NYX is designed to augment human operators, not replace them. Its purpose is to improve operational scalability and decision quality while ensuring organizations remain in control of how automation is applied within their environment.

SECURITY & GOVERNANCE

NYX is designed with a strong emphasis on operational trust, auditability, governance, and controlled decision transparency.

As organizations increasingly evaluate automation and AI-assisted security workflows, governance and explainability become critical operational requirements rather than optional capabilities. NYX was intentionally designed to support these requirements through structured decisioning, configurable operational controls, and auditable workflow visibility.

Rather than functioning as a black-box automation platform, NYX is architected to provide transparent, traceable, and governable operational decision workflows across enterprise environments.

Explainable Decisioning & Operational Transparency

At the core of NYX is the Trinity Decision Model, which introduces structured consensus-based reasoning into the operational decision lifecycle.

Each decision generated by NYX may include:

- Analyst evaluation outputs
- Judge validation logic
- Auditor challenge and verification results
- Decision confidence indicators
- Contextual enrichment references
- Operational timestamps and TTD metrics

- Human approval or override records where applicable

This model enables organizations to maintain visibility into how operational decisions were reached, what contextual intelligence influenced those outcomes, and how governance policies were applied throughout the workflow.

By preserving operational decision lineage and audit traceability, NYX supports more defensible and transparent security operations.

Auditability & Decision Traceability

NYX is designed to generate comprehensive audit records surrounding operational decisions, workflow actions, and automation events.

Audit records may include:

- Alert and event metadata
- Correlated enrichment data
- Decision outputs
- Workflow actions taken
- Approval and escalation history
- Automation execution records
- User interactions and overrides
- Operational timing metrics

This audit model enables organizations to:

- Support governance and compliance initiatives
- Improve operational accountability
- Validate workflow consistency
- Assist incident review and forensic analysis
- Enhance executive and operational reporting

Access Control & Operational Governance

NYX supports role-based access control (RBAC) and governed operational permissions designed to align with enterprise security and segregation-of-duty requirements.

Organizations may configure:

- Administrative roles
- Analyst review permissions
- Approval authorities

This governance model enables organizations to maintain operational control while progressively scaling automation maturity in a controlled and auditable manner.

Data Security & Operational Control

NYX is architected primarily for controlled-environment and on-premise deployment models where data governance, infrastructure segmentation, and operational visibility requirements are critical.

Security-focused design considerations may include:

- Local operational data processing
- Configurable retention policies
- Encryption of operational data and credentials
- Segregation of operational workflows
- Restricted external dependency requirements
- Air-gapped deployment support
- Immutable or protected audit logging architectures where applicable

This approach enables organizations to maintain greater control over operational data, decision workflows, and infrastructure governance requirements.

Operational Trust by Design

NYX was designed with the understanding that advanced operational decisioning systems must remain governable, explainable, and controllable within enterprise environments.

The platform intentionally avoids reliance on opaque autonomous workflows by emphasizing:

- Structured consensus-based reasoning
- Human-governed automation
- Policy-driven operational controls
- Transparent decision workflows
- Auditable operational outputs

This governance-first architecture enables organizations to pursue operational scalability and accelerated decisioning without sacrificing trust, visibility, or oversight.

STRATEGIC OUTLOOK

Modern cybersecurity operations are approaching a critical inflection point. As enterprise environments continue expanding across cloud, identity, infrastructure, endpoint, and hybrid operational domains, the volume and complexity of security telemetry will continue to grow faster than traditional analyst-driven workflows can realistically scale.

Detection capabilities will continue to improve. Telemetry volume will continue to increase. Automation adoption will continue to accelerate. But the defining challenge for modern security operations will increasingly become operational decisioning at scale.

Organizations must be able to rapidly determine:

- What matters
- What requires action
- What can be deprioritized
- What can be automated safely
- What decisions can be operationally defended

The future of cybersecurity operations will not be defined solely by detection capability, but by the ability to consistently produce structured, explainable, and governable operational decisions across increasingly complex environments.

NYX was designed specifically to address this challenge.

Rather than replacing existing security technologies or human operators, NYX introduces a parallel operational intelligence and decision layer that enhances existing workflows through contextual enrichment, consensus-based reasoning, and human-governed operational automation.

By combining:

- Cross-domain operational visibility
- Structured decisioning
- Trinity-based consensus evaluation
- Configurable governance controls
- Human review workflows
- Operationally aligned automation

NYX enables organizations to improve scalability, reduce analyst overload, accelerate Time to Decision (TTD), and strengthen operational consistency without sacrificing governance, transparency, or human oversight.

Operational decisioning will become one of the defining cybersecurity challenges of the next decade.

NYX is designed to help organizations meet that challenge.

Different by design.

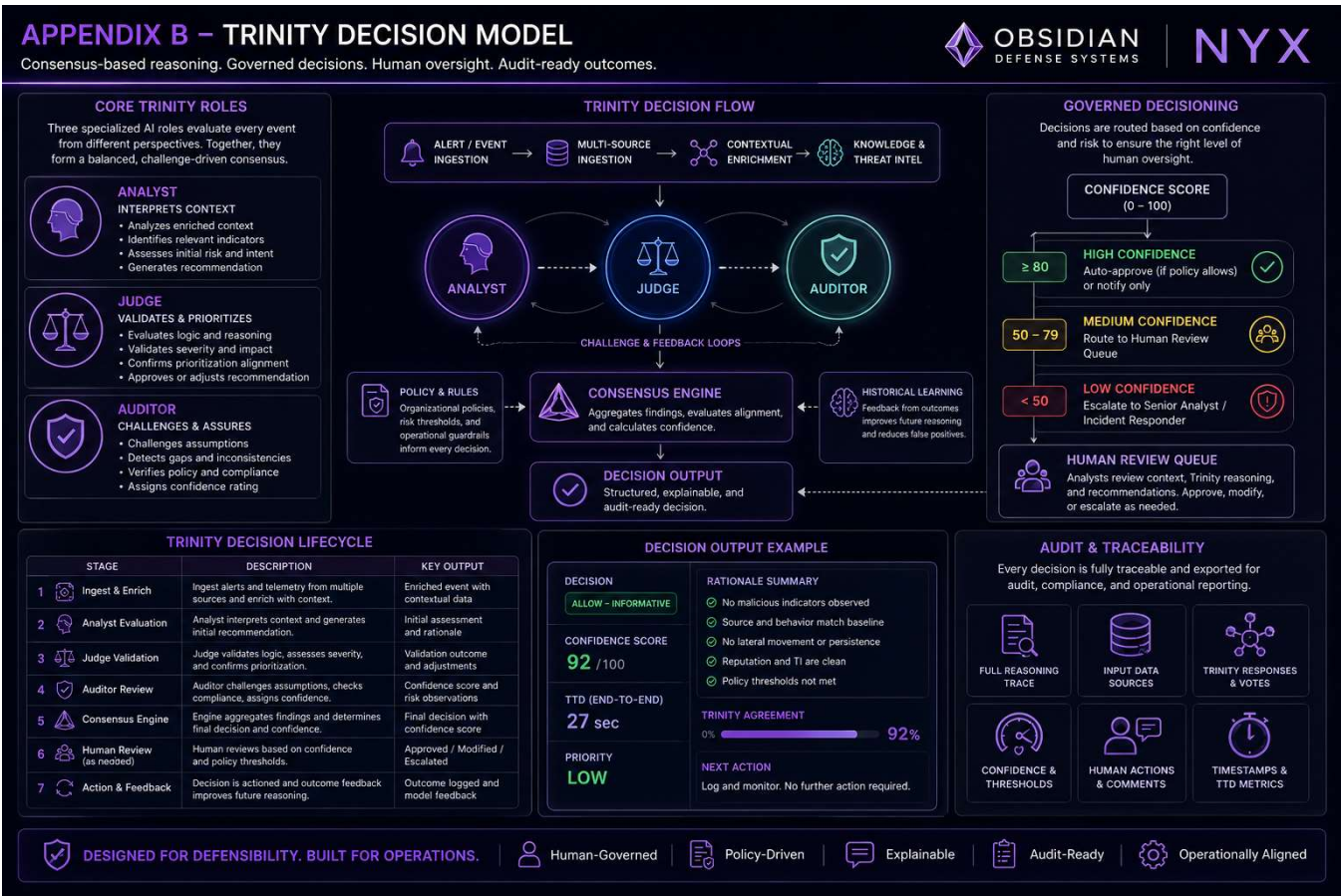
Direction is clear

APPENDIX

APPENDIX A – NYX PLATFORM DIFFERENTIATION

APPENDIX A – PLATFORM DIFFERENTIATION					
NYX is purpose-built for operational decisioning at scale, not just detection, automation, or event management.					
CAPABILITY	SIEM	SOAR	AIOps / ITOM	COPILOT / AI ASSISTANT	NYX OPERATIONAL DECISIONING PLATFORM
 Primary Purpose	Detect and aggregate security events	Orchestrate and automate security workflows	Optimize IT operations and reduce noise	Assist users with information and guidance	Produce structured, defensible operational decisions at scale
 Core Function	Collect, normalize, and correlate events	Automate tasks and orchestrate workflows	Identify anomalies and optimize operations	Provide conversational assistance and recommendations	Ingest, enrich, evaluate, and decide
 Correlation	Primary Capability	Limited (relies on SIEM / sources)	Primary Capability (IT focused)	Limited / Indirect	Core Capability across multiple domains
 Automation	Limited (alerts / enrichment rules)	Primary Capability	Limited (remediation not security-focused)	Limited / User-initiated	Governed, policy-driven (optional and configurable)
 Contextual Enrichment	Limited (via integrations)	Limited (via playbooks and integrations)	IT context focused	Limited	Core Capability across multiple security domains
 Human Review & Approval	Manual process outside the platform	Optional (workflow based)	Limited	User driven	Built-in, configurable human review workflows
 Consensus Decisioning (Analyst, Judge, Auditor)	Not Available	Not Available	Not Available	Not Available	Core via Trinity Decision Model
 Operational Decisioning	Not Available (Provides alerts)	Partial (Executes actions based on inputs)	Indirect	Not Available	Primary Purpose
 Auditability & Traceability	Log based (event centric)	Workflow logs	Log based	Limited	Full decision traceability with explainability and reasoning
 Cross-Domain Reasoning	Limited	Limited	Limited (IT domains)	Limited	Core Capability (security and infrastructure domains)
 Governance Controls	Role-based access	Role & workflow controls	IT operations controls	Basic access controls	Granular governance, RBAC, policy, and decision controls
 TTD Visibility	Not Available	Not Available	Not Available	Not Available	End-to-End TTD Metrics and Visibility
 Deployment Flexibility	On-prem / Cloud / Hybrid	On-prem / Cloud / Hybrid	On-prem / Cloud / Hybrid	Cloud / SaaS	On-prem / Cloud / Hybrid / Air-gapped
 Explainability	Limited (rule based)	Limited (workflow based)	Limited	Limited / Varies	Built-in explainability with reasoning and confidence
 Decision Traceability	Event trace only	Workflow execution trace	Log trace	Limited	Complete decision lifecycle trace (ingest → decision → action)

APPENDIX B – TRINITY DECISION MODEL



APPENDIX F – EXAMPLE INTEGRATION ARCHITECTURE

